

SYSTEM & UML DIAGRAMS

AI Chatbot for College · BBA Computer Applications

1. Use Case Diagram

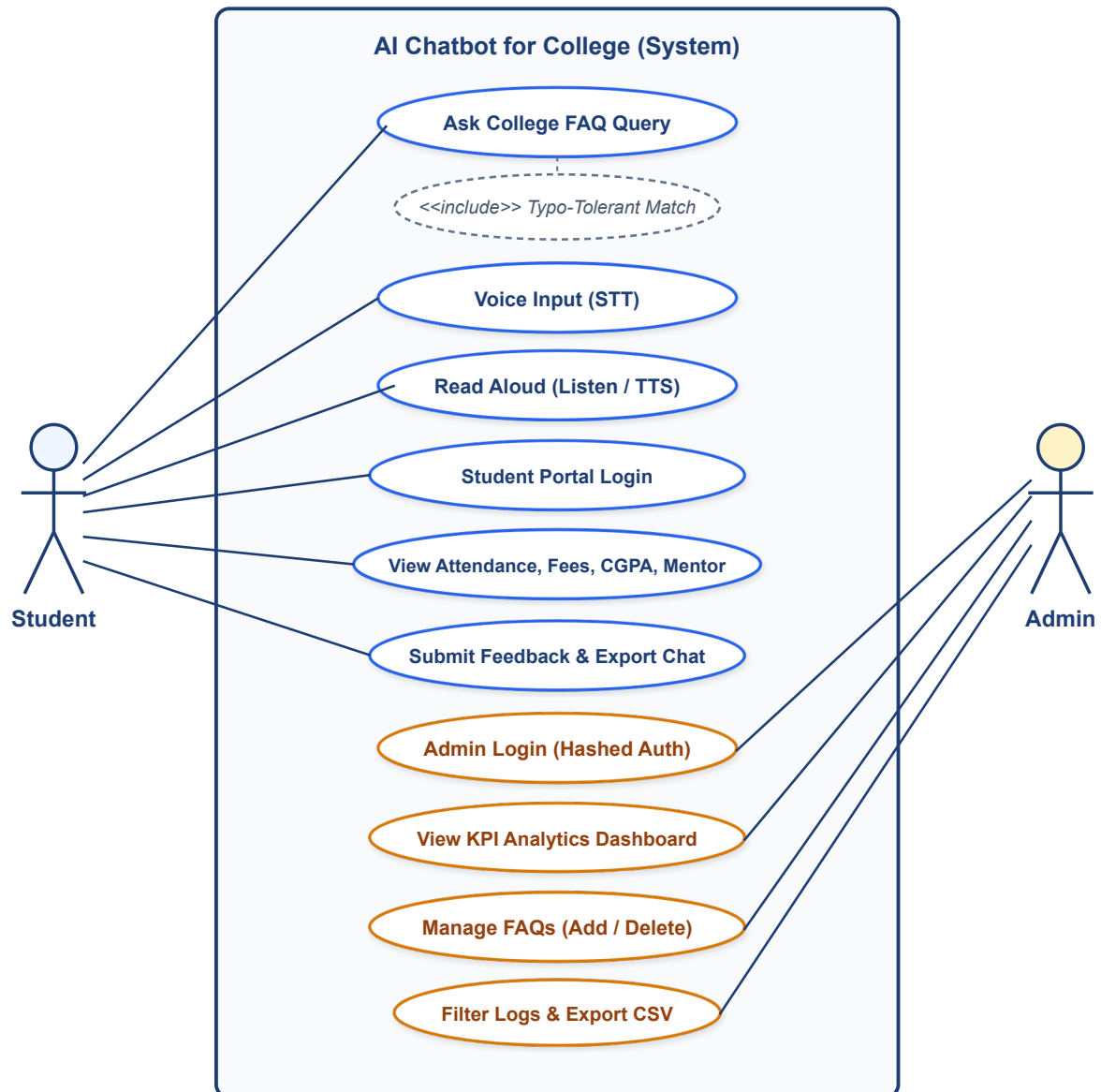


Fig 1: Use Case Diagram — Student and Admin actors and their interactions with the system.

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2. Class Diagram (With Full Relations & Multiplicities)

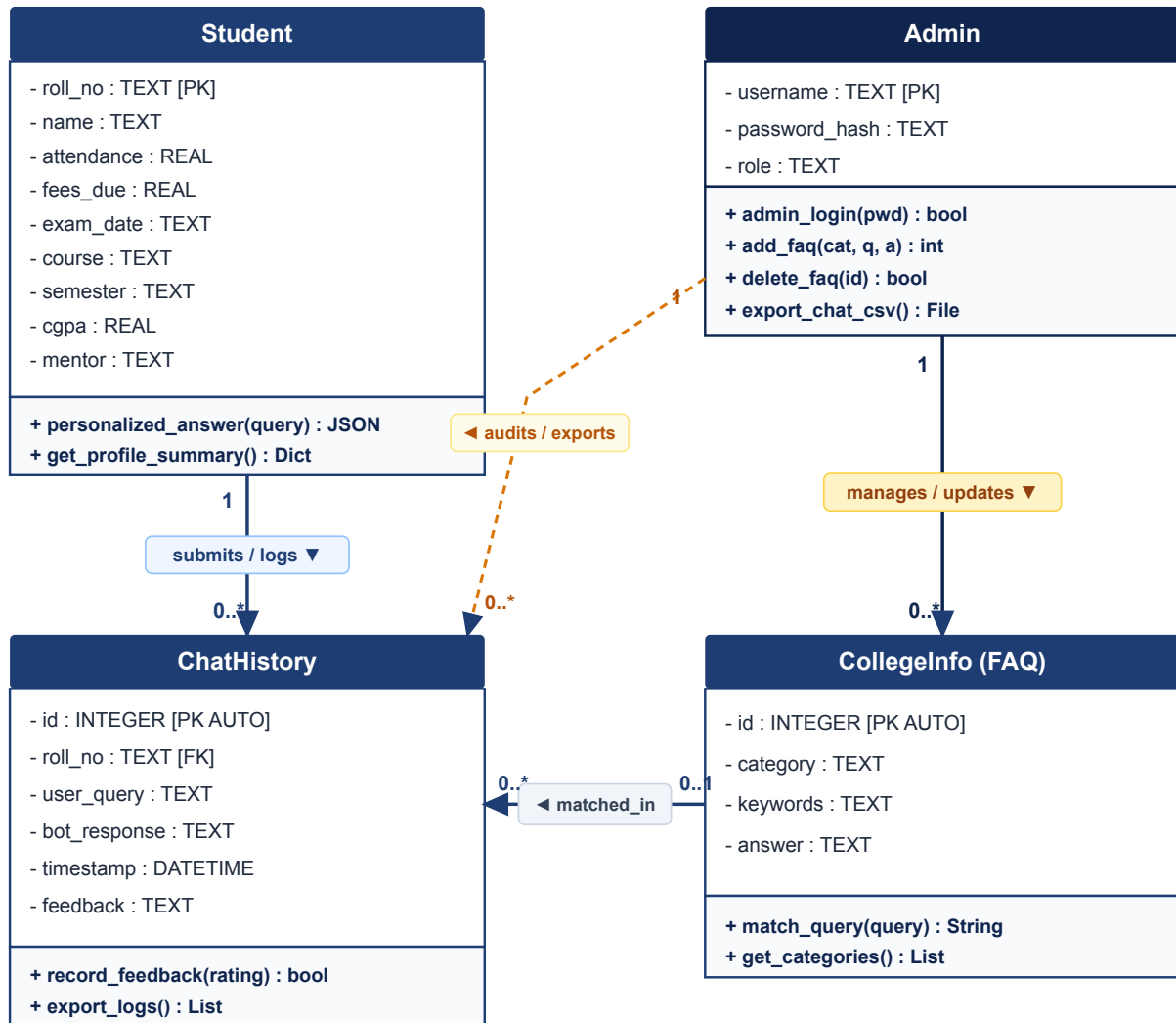


Fig 2: Class Diagram — fully connected system classes with associations, multiplicities (1 to 0..*), and role verbs.

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3. Object Diagram (Runtime Instances & Slot Values)

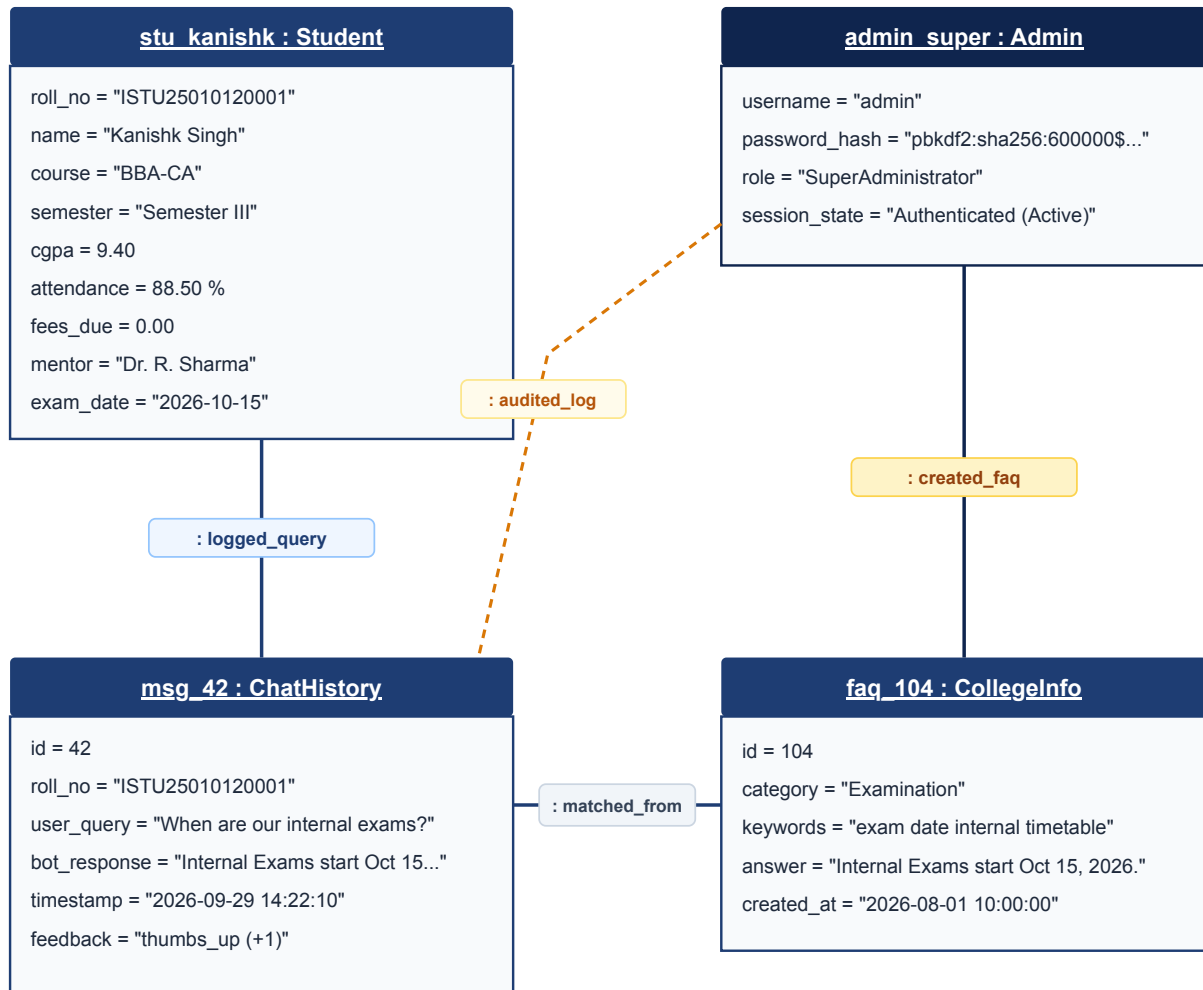


Fig 3: Object Diagram — concrete runtime object instances with slot values and links during an active student query session.

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4. Sequence Diagram — Query, Feedback & Session Termination

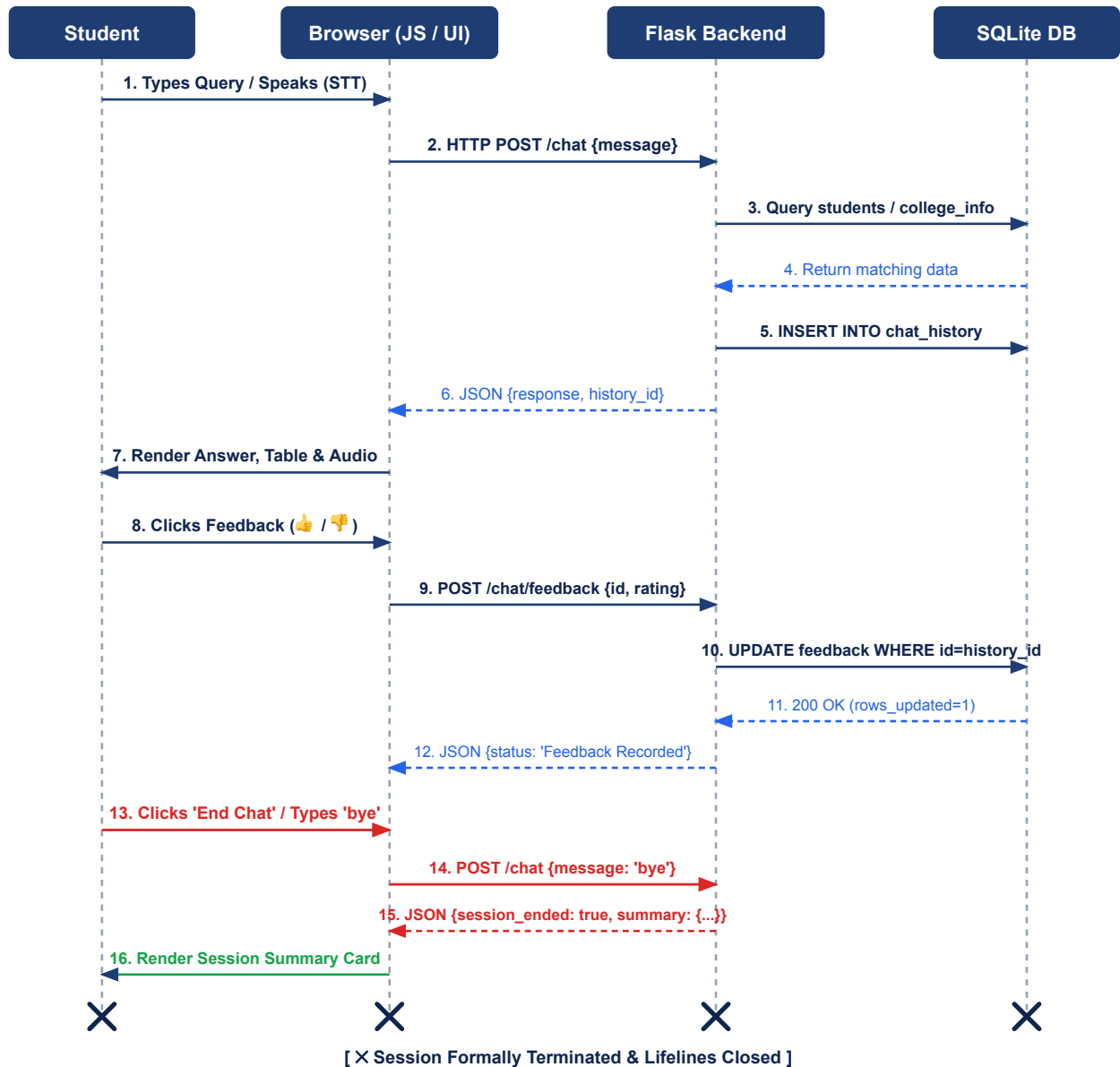


Fig 4: Sequence Diagram — complete query, feedback ACK, explicit End Chat flow, and standard UML lifeline termination (X).

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5. Activity Diagram — Chatbot Response & Session Flow

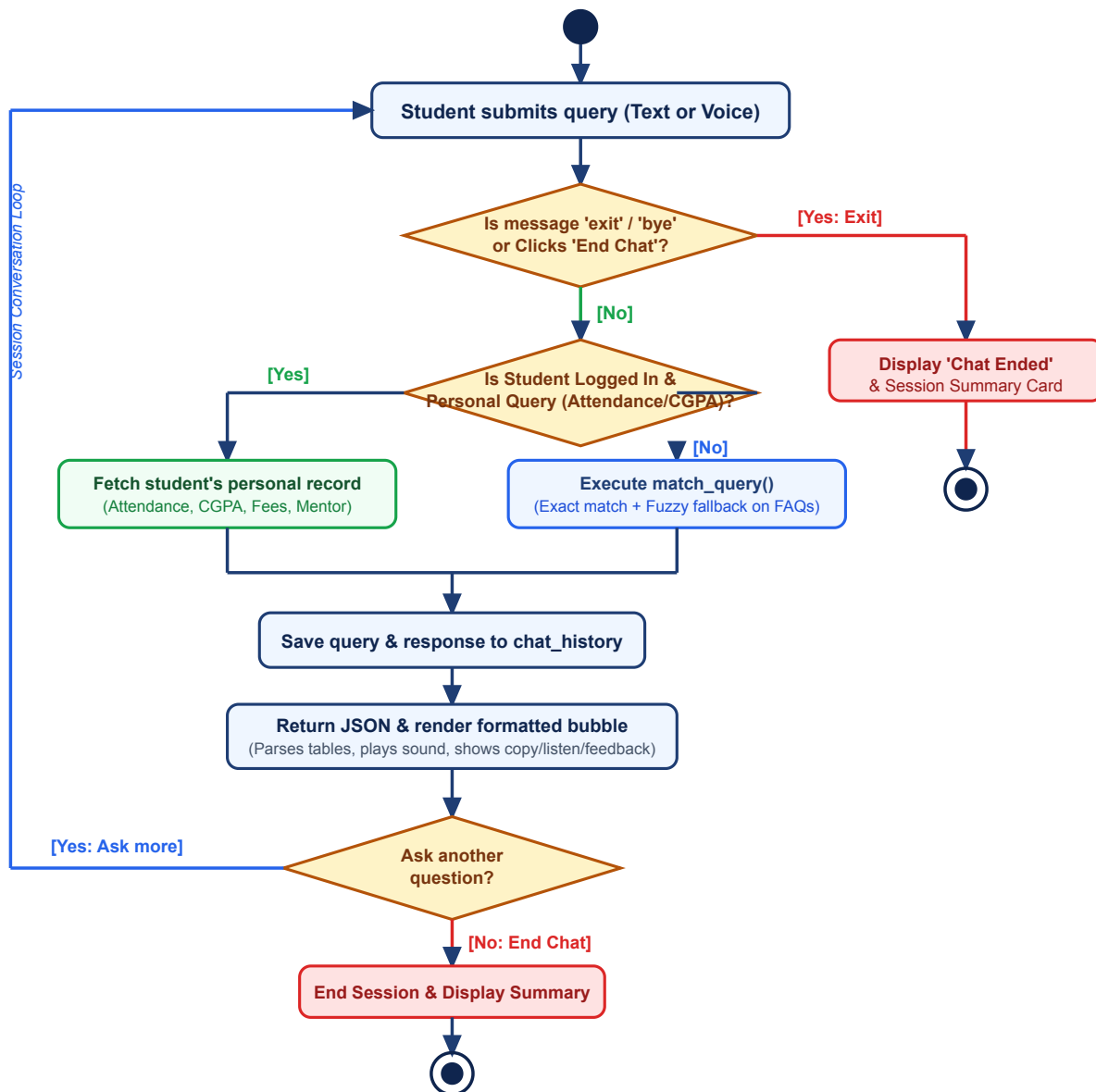


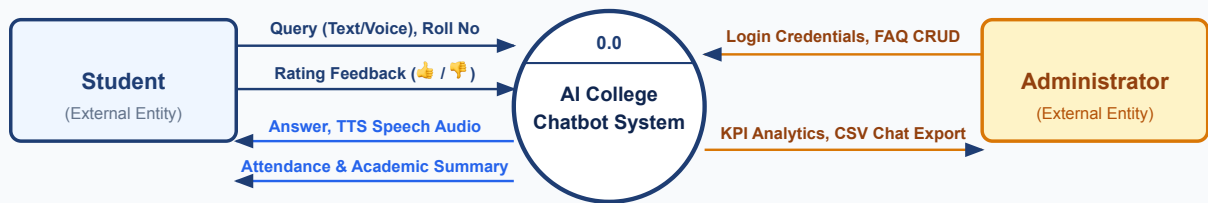
Fig 5: Activity Diagram — decision flow from receiving a message to displaying the response, showing the controlled session loop and explicit End state.

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6. Data Flow Diagram (DFD Level 0 Context & Level 1 Detailed)

DFD Level 0: Context Diagram



DFD Level 1: Process Decomposition

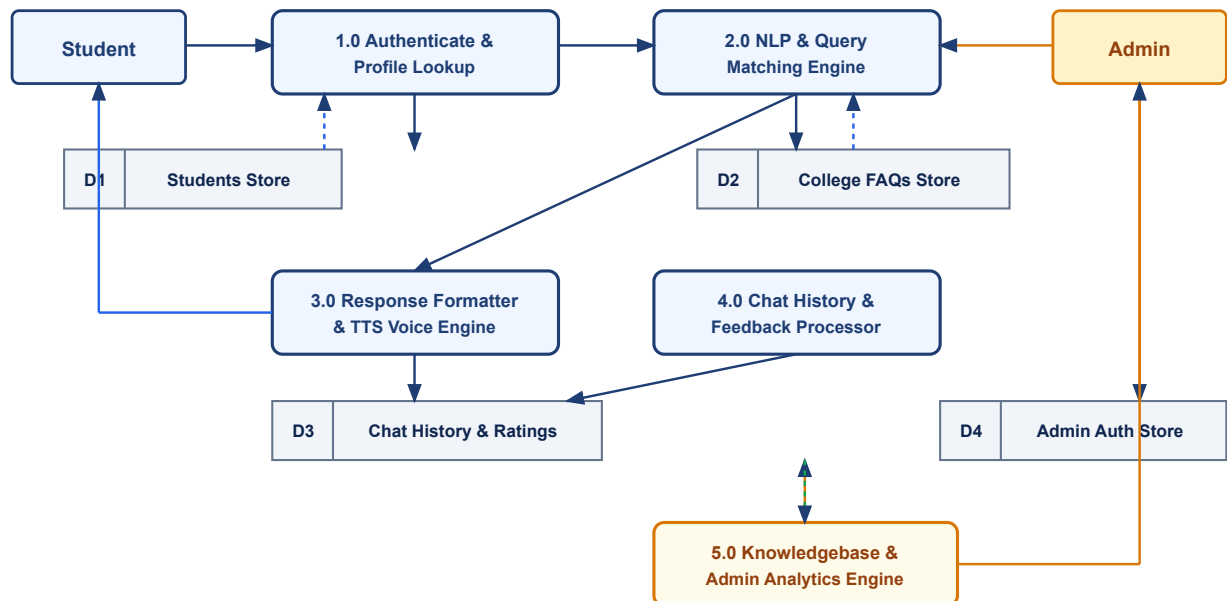


Fig 6: Data Flow Diagram — Level 0 (Context Diagram) and Level 1 (Process Decomposition with Data Stores D1–D4).

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7. Entity-Relationship Diagram (ERD — Database Architecture)

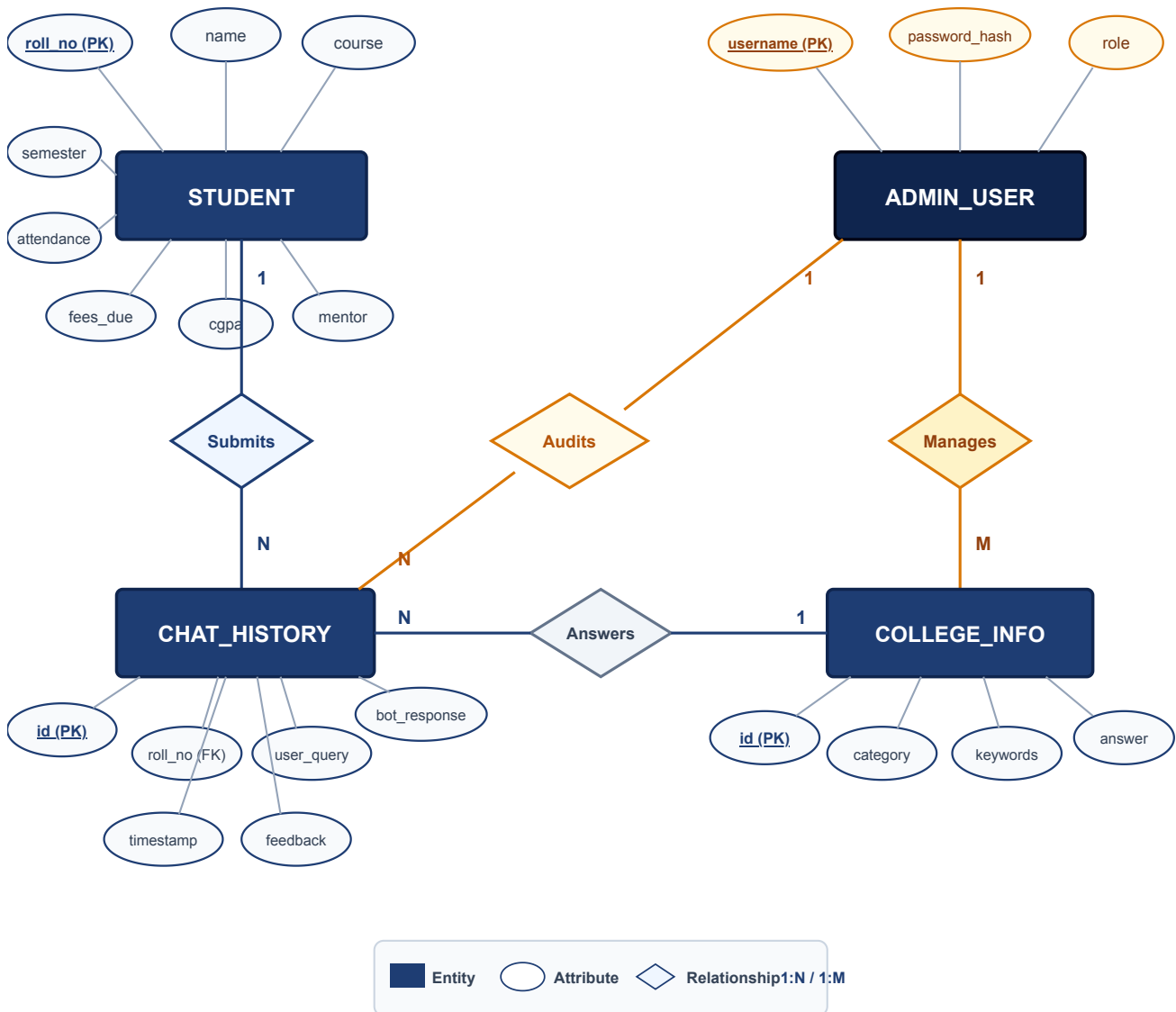


Fig 7: Entity-Relationship Diagram — SQLite database schema with Primary Keys (PK), Foreign Keys (FK), attributes, and cardinality mapping.

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8. Deployment Diagram (Hosting Topology & Artifact Architecture)

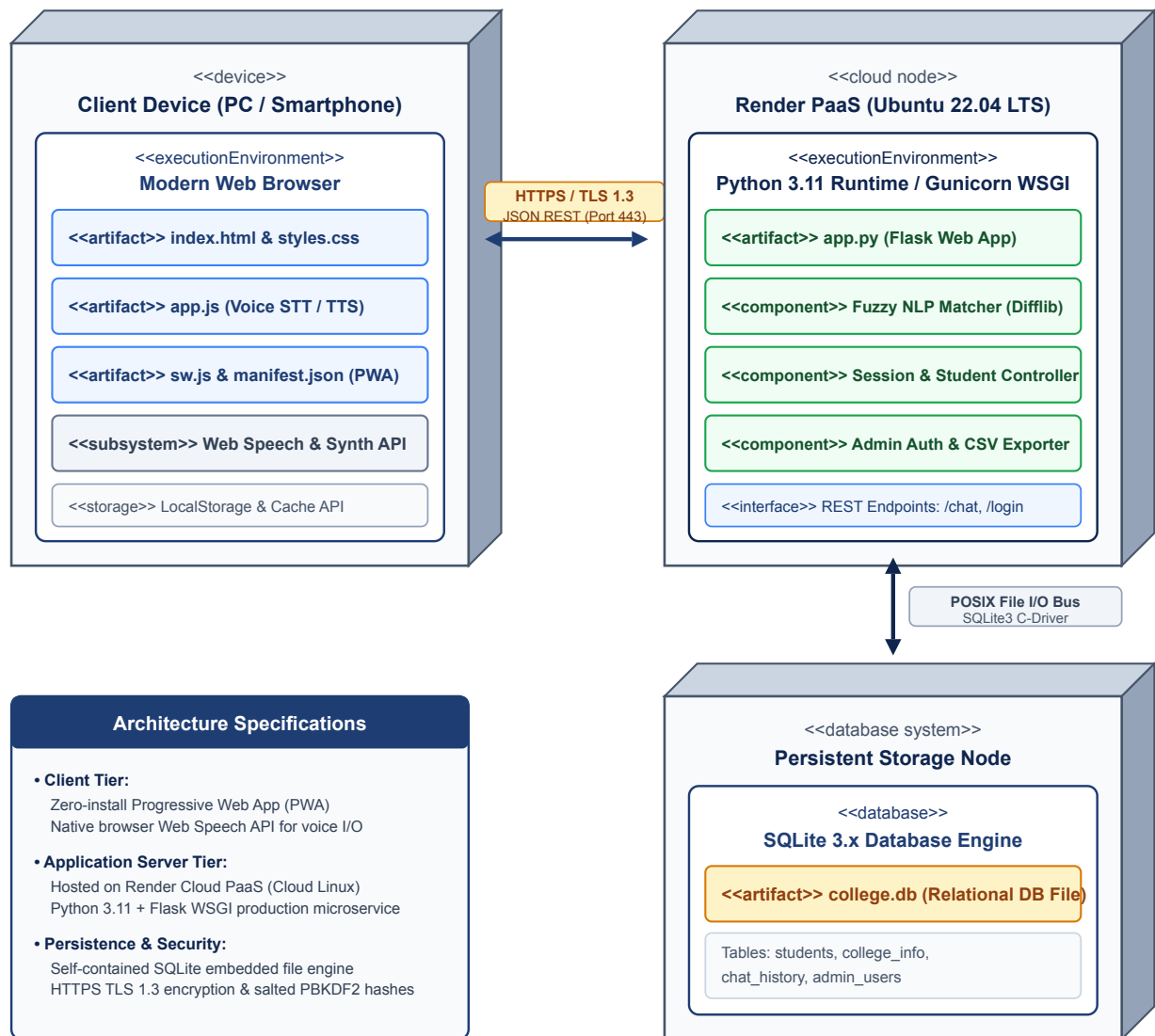


Fig 8: Deployment Diagram — client device, cloud hosting tier (Render Linux), embedded SQLite persistent storage, and protocol architecture.